

COMP0083: Kernel Methods

Lecture Notes

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1 Introduction to Kernels

1. Kernels let us compare objects on the basis of features
2. Kernel methods can control smoothness and avoid overfitting/underfitting

2 Inner Product

Let H be a vector space over $\mathbb{R}$. A function $\langle \cdot, \cdot \rangle_H : H \times H \rightarrow \mathbb{R}$ is an inner product if:

- (i) **Linear:** $\langle \alpha f, \beta g \rangle_H = \alpha \langle f, g \rangle_H + \alpha_2 \langle f_2, g \rangle_H$
- (ii) **Symmetric:** $\langle f, g \rangle_H = \langle g, f \rangle_H$
- (iii) $\langle f, f \rangle_H \geq 0$ and $\langle f, f \rangle_H = 0 \Leftrightarrow f = 0$ (positive definiteness)

$$\|f\|_H := \sqrt{\langle f, f \rangle_H}$$

3 Inner Product on $\mathbb{R}^n$

A function $\langle \cdot, \cdot \rangle : \mathbb{R}^n \times \mathbb{R}^n \rightarrow \mathbb{R}$ is an inner product on $\mathbb{R}^n$ if:

$\exists$ symmetric positive definite matrix M s.t. $\langle x, y \rangle = x^T M y$ for $\forall x, y \in \mathbb{R}^n$

For our product with $M = I$:

$$\langle x, y \rangle = x^T y$$

4 Hilbert Space

A Hilbert space is an inner product space containing all Cauchy sequence limits.

5 Kernel Definition

Let X be a non-empty set. A function $k : X \times X \rightarrow \mathbb{R}$ is a kernel if there exists a Hilbert space H and a feature map $\phi : X \rightarrow H$ such that:

$$k(x, x') = \langle \phi(x), \phi(x') \rangle_H$$

6 Key Properties of Kernels

1. A single kernel can correspond to several possible features.
2. **Sums of kernels are kernels:** Given $\alpha > 0$, k, k_2, k_3 are kernels on X , then αk and $k_1 + k_2$ are kernels on X .

3. A difference of kernels may not be a kernel.
4. **Mappings between spaces:** Let X and $\tilde{X}$ be sets and a map $A : X \rightarrow \tilde{X}$. Define kernel on $\tilde{X}$. Then $k(A(x), A(x'))$ is a kernel on X .
5. **Products of kernels are kernels:** Given k_1 on X_1 and k_2 on X_2 , then $k_1 \times k_2$ is a kernel on $X_1 \times X_2$. If $X_1 = k_1 = X$, then $k := k_1 \times k_2$ is a kernel on X .

7 Frobenius Inner Product

$$\langle A, B \rangle_F = \sum_{ij} A_{ij} B_{ij} = \text{tr}(A^T B), \quad A^* \text{ is conjugate-transpose of } A$$

8 Polynomial Kernel

$$k(x, x') := (\langle x, x' \rangle + c)^m$$

where m must be an integer and $m \geq 1, c \geq 0$.

9 Convergence Radius

For $r \in (0, \infty]$, with $a_n \geq 0$ for all $n \geq 0$:

$$f(z) = \sum_{n=0}^{\infty} a_n z^n, \quad |z| < r, \quad z \in \mathbb{R}$$

Define X_0 to be the $\sqrt{r}$ -ball in $\mathbb{R}^d$, so $\|x\| < \sqrt{r}$:

$$k(x, x') = f(\langle x, x' \rangle) = \sum_{n=0}^{\infty} a_n \langle x, x' \rangle^n$$

10 Sequence Spaces

The space ℓ_2 comprises all sequences $a := \{a_i\}_{i \geq 1}$ for which:

$$\|a\|_{\ell_2}^2 = \sum_{i=1}^{\infty} a_i^2 < \infty$$

11 Properties of Kernels and Positive Definiteness

1. Kernels are positive definite. Reverse also holds.

11.1 Norm

Let F be a vector space over the field $\mathbb{R}$ (or $\mathbb{C}$). A function $\|\cdot\|_F : F \rightarrow [0, \infty)$ is said to be a norm if:

- (i) $\|f\|_F = 0 \Leftrightarrow f = 0$ (norm separates points)
- (ii) $\|\lambda f\|_F = |\lambda| \|f\|_F, \forall \lambda \in \mathbb{R}, \forall f \in F$ (positive homogeneity)
- (iii) $\|f + g\|_F \leq \|f\|_F + \|g\|_F, \forall f, g \in F$ (triangle inequality)

In every normed vector space, one can define a metric induced by the norm: $d(f, g) = \|f - g\|_F$.

In every inner product vector space, one can define a norm induced by the inner product:

$$\|f\|_F = \langle f, f \rangle_F^{1/2}$$

11.2 Orthogonal Complement

The orthogonal complement of M : $M^\perp = \{g \in F : \langle f, g \rangle = 0, \forall f \in M\}$
 $M^\perp$ is a linear subspace of F ; $M \cap M^\perp = \{0\}$.

11.3 Key Relations in Inner Product Space

1. **Cauchy-Schwarz inequality:** $|\langle f, g \rangle| \leq \|f\|_F \|g\|_F$
2. **Parallelogram law:** $2\|f\|_F^2 + 2\|g\|_F^2 = \|f + g\|_F^2 + \|f - g\|_F^2$
3. **Polarization identity:** $4\langle f, g \rangle = \|f + g\|_F^2 - \|f - g\|_F^2$
4. **Pythagorean theorem:** $f \perp g \Rightarrow \|f\|_F^2 + \|g\|_F^2 = \|f + g\|_F^2$

11.4 Convergence and Cauchy Sequences

From $\|f_n - f_m\|_F \leq \|f_n - f\|_F + \|f - f_m\|_F$, we get: Convergent $\Rightarrow$ Cauchy, Cauchy $\Rightarrow$ Convergent?

Example: $1, 1.4, 1.41, 1.414, 1.4142, \dots$ is Cauchy in $\mathbb{Q}$ which does not converge because $\sqrt{2} \notin \mathbb{Q}$.

11.5 Complete Metric Spaces

A metric space F is said to be complete if every Cauchy sequence $\{f_n\}_{n=1}^\infty$ in F converges: it has limit and this limit is in F , i.e., one can find $f \in F$ s.t. $\lim \|f_n - f\|_F = 0$.

Complete + Norm = Banach Complete + Inner product = Hilbert

11.6 Closed vs. Complete

Closed v.s. Complete: Closed: $M \subseteq F$ is closed in $F \Leftrightarrow$ it contains limits of all sequences in M that converge in F .

Complete: M is complete if all Cauchy sequence in M converge in M .

11.7 Linear Operators

Linear operator: $A : F \rightarrow G$, where F and G are both vector spaces over $\mathbb{R}$. A is said to be a linear operator if:

$$A(\alpha_1 f_1 + \alpha_2 f_2) = \alpha_1 A(f_1) + \alpha_2 A(f_2), \quad \forall \alpha_1, \alpha_2 \in \mathbb{R}, \quad f_1, f_2 \in F$$

Operators with $G = \mathbb{R}$ are called **functionals**.

11.8 Operator Norm

Operator norm: The operator norm of a linear operator $A : F \rightarrow G$ is defined as:

$$\|A\| = \sup_{f \in F} \frac{\|Af\|_G}{\|f\|_F}$$

If $\|A\| < \infty$, A is called a **bounded linear operator**. $\|A\|$ is the smallest number λ s.t. the inequality $\|Af\|_G \leq \lambda \|f\|_F$ holds for all $f \in F$.

11.9 Continuity

Continuity: $A : F \rightarrow G$, where F and G are both normed vector spaces over $\mathbb{R}$. A is said to be continuous at $f_0 \in F$, if $\forall \varepsilon \geq 0, \exists \delta = S(\varepsilon, f_0) > 0$, s.t.

$$\|f - f_0\|_F < \delta \Rightarrow \|Af - Af_0\|_G < \varepsilon$$

A is said to be continuous on F if it is continuous at every point of F .

Example: For $g \in F$, $A_g : F \rightarrow \mathbb{R}$ defined with $A_g(f) = \langle f, g \rangle_F$ is continuous on F .

$|A_g f_1 - A_g f_2| = |\langle f_1 - f_2, g \rangle| \leq \|g\|_F \|f_1 - f_2\|_F$, so take $\delta = \frac{\varepsilon}{\|g\|_F}$ (also Lipschitz?)

12 Continuous and Bounded Operators

1. **Continuous function:** $A : F \rightarrow G$ maps open (closed) sets from open (closed) sets.
2. **Bounded linear operator:** $A : F \rightarrow G$ maps bounded sets to bounded sets.
3. **Continuous operator = Bounded operator:** Let $(F, \|\cdot\|_F)$ and $(G, \|\cdot\|_G)$ be normed linear spaces. If L is a linear operator, then the following three conditions are equivalent:
 - (i) L is bounded operator
 - (ii) L is continuous on F
 - (iii) L is continuous at one point of F

13 Riesz Representation Theorem

Let H be a Hilbert space and let $f : H \rightarrow \mathbb{F}$ be a bounded linear functional. Then $\exists! y \in H$ s.t.

$$f(x) = \langle x, y \rangle_H \quad \forall x \in H \quad \text{and} \quad \|f\| = \|y\|_H$$

Every continuous linear score is just a dot-product with some hidden vector.

$f \in H'$ (continuous dual of H), $\tilde{F} := \{f : H \rightarrow \mathbb{F} \mid f \text{ is linear and } \|f\|_{H'} = \sup \frac{\|f\|}{\|x\|_H}\}$.

The Riesz isomorphism is the mapping $J : H \rightarrow H'$, $J(y)(x) = \langle x, y \rangle_H$, and it is bijective and norm-preserving ($\|f\|_H = \|y\|_H$), so it is isometric isomorphism.

14 Evaluation Functional

Given a space of functions X and a fixed point t_0 in the domain, the evaluation functional at $t_0 : E_{t_0} : X \rightarrow \mathbb{F}$, $E_{t_0}(f) = f(t_0)$.

It is linear but not always bounded.

15 Reproducing Kernel Hilbert Space

Reproducing kernel Hilbert Space $\Leftrightarrow$ every evaluation functional $E_x : H \rightarrow \mathbb{F}$, $E_x(f) = f(x)$, is bounded with respect to norm of H .

Bounded evaluations $\Rightarrow$ Reproducing kernel: Because $E_x \in H'$ is bounded, from Riesz, $\exists k(\cdot, x) = k_x \in H$ s.t. $f(x) = E_x(f) = \langle f, k(\cdot, x) \rangle_H \forall f \in H$.

The function $k : X \times X \rightarrow \mathbb{F}$, $k(x, y) := k_y(x) = \langle k_y, k_x \rangle_H$, is called **reproducing kernel**.

Property: $f(x) = \langle f, k(\cdot, x) \rangle_H$ is the **reproducing property**.

Reproducing kernel $\Rightarrow$ bounded evaluation.

15.1 Moore-Aronszajn Theorem

If k is positive-definite on $X \times X$, $\exists!$ RKHS H_k where kernel is k s.t.

$$f(x) = \langle f, k(\cdot, x) \rangle_{H_k}, \quad \forall f \in H_k, \quad x \in X$$

$|f(x)| \leq \|f\|_{H_k} \|k(\cdot, x)\|_{H_k}$ and hence E_x is bounded.

16 Properties of Reproducing Kernels

1. **Reproducing kernel is unique and RKHS:** It is a reproducing kernel Hilbert space $\Leftrightarrow$ it has reproducing kernel.
2. **Every reproducing kernel is a kernel:** We can take $\phi : x \mapsto k(\cdot, x)$, $k(x, y) = \langle k(\cdot, x), k(\cdot, y) \rangle_{H_k}$, i.e., $\mathbb{R}HSH$ is a feature space.

17 Orthonormal Basis

In Hilbert space H , an orthonormal basis is a family $\{e_k\}_{k \in B}$ of elements of H satisfying:

- (i) **Orthogonality:** $\langle e_i, e_j \rangle = 0 \ \forall i, j \in B$ with $i \neq j$
- (ii) **Normalization:** $\|e_k\|_H = 1$ for $k \in B$
- (iii) **Completeness:** linear span of family $\{e_k, k \in B\}$ is dense in H

18 Spectral Theorem

Let X be a compact metric space and $k : X \times X \rightarrow \mathbb{R}$ a continuous kernel. Define

$$H = \left\{ f = \sum_{j \in J} a_j e_j : \left\{ \frac{a_j}{\sqrt{\lambda_j}} \right\} \in \ell^2(J) \right\} \quad \text{with inner product:}$$

$$\left\langle \sum_{j \in J} a_j e_j, \sum_{b \in J} b_j e_j \right\rangle_H = \sum_{j \in J} \frac{a_j b_j}{\lambda_j}$$

Then $H = H_k$.

19 Spectral Decomposition

Let F be a Hilbert space, and $T : F \rightarrow F$ a compact, self-adjoint operator. There is an at most countable orthonormal set (CONS) $\{e_j\}_{j \in J}$ of F and $\{\lambda_j\}_{j \in J}$ with $|\lambda_1| > |\lambda_2| > \dots > 0$ converging to 0, such that

$$f = \sum_{j \in J} \langle f, e_j \rangle e_j \quad \text{and} \quad Tf = \sum_{j \in J} \lambda_j \langle f, e_j \rangle e_j, \quad f \in F$$

20 Integral Linear Operator

$$(Kf)(x) = \int_X f(y) K(x, y) dy$$

It is linear because $[K(\alpha f_1 + \beta f_2)](x) = \alpha[Kf_1](x) + \beta[Kf_2](x)$.

21 Mercer's Theorem

Let k be a continuous kernel on compact metric space X , and let ν be a finite Borel measure on X with $\text{supp}(\nu) = X$. Then $\forall x, y \in X$:

$$k(x, y) = \sum_{j \in J} \lambda_j e_j(x) e_j(y)$$

$\{\lambda_j e_j\}$ and $\{e_j\}_{j \in J}$ are eigenvalues and eigenfunctions of $T_{k, \nu}$ on the RKHS, and the convergence of sum is uniform on $X \times X$ and absolute for each pair $(x, y) \in X \times X$.

$$k(x, y) = \sum_{j \in J} \lambda_j e_j(x) e_j(y) = \langle \sqrt{\lambda_j} e_j(x), \sqrt{\lambda_j} e_j(y) \rangle_u$$

22 Alternative Formulation

Item 43 can be viewed in a different way:

Rescale the eigenfunctions: $\varphi_j := \sqrt{\lambda_j} e_j$ and $\langle \varphi_i, \varphi_j \rangle_H = \delta_{ij}$, so $\langle e_i, e_j \rangle_H = \frac{\delta_{ij}}{\lambda_j}$.

M is Gram matrix of the original eigenfunctions $\{e_j\}$ in the RKHS.

So any function $f(x) = \sum f_n e_n(x) = \sum \frac{\hat{f}_n}{\sqrt{\lambda_n}} \varphi_n(x)$

$$\|f\|_H^2 = \sum \frac{|\hat{f}_n|^2}{\lambda_n}$$

For the evaluation functional $E_x(f) = f(x)$, Cauchy-Schwarz gives:

$$|f(x)|^2 = |E_x(f)|^2 \leq \left(\sum \frac{\hat{f}_n^2}{\lambda_n} \right) \left(\sum \|e_n\|_H^2 k(x, x) \right) = \|f\|_H^2 k(x, x)$$

For E_x bounded, we need $\left\{ \frac{|f_n|}{\sqrt{\lambda_n}} \right\} \in \ell^2(J)$ and $k(x, x) < \infty$.

For other ordinary dot product in L_2 , $\lambda_n = 1 \Rightarrow$ the series diverges $\Rightarrow$ unbounded $\Rightarrow$ not RKHS.

23 Infinite Feature Space via Fourier Series

$$k(x, y) = \sum_{j \in J} \lambda_j e_j(x) e_j(y) = \sum_{j \in J} \lambda_j \exp(i\omega_j(x - y))$$

which is translation invariant.

24 Eigenvalues and Eigenfunctions

Since $\{\lambda_j\}_{j \in J}$ and $\{e_j\}_{j \in J}$ are eigenvalues and eigenfunctions of T_k , then we have:

$$T_k e_j(x) = \int e_j(y) k(x, y) \nu(y) dy = \int e_j(y) \sum_{j'} \lambda_{j'} e_{j'}(x) e_{j'}(y) \nu(y) dy = \lambda_j e_j(x)$$

25 Small RKHS Norm and Smooth Functions

Small RKHS norm results in smooth functions.

25.1 Two Functions Identical in RKHS Norm

Two functions identical in RKHS norm agree at every point:

$$|f(x) - g(x)| = |E_x(f - g)| = |\langle f - g, k_x \rangle_H| \leq \|f - g\|_H \|k_x\|_H = 0$$

$$\|k_x\|_H = \langle k(\cdot, x), k(\cdot, x) \rangle_H^{\frac{1}{2}} = k(x, x)^{\frac{1}{2}}$$

25.2 Distance Between Means in Feature Space

$$\left\| \frac{1}{m} \sum_{i=1}^m \phi(x_i) - \frac{1}{n} \sum_{j=1}^n \phi(y_j) \right\|_H^2 \rightarrow \text{distance between their means in feature space}$$

When $\phi(x) = x$, distinguish means. When $\phi(x) = Lxx^2$, distinguish means and variances.

26 Principal Component Analysis (PCA)

PCA: First principal component (max. variance)

$$u_1 = \arg \max_{\|u\|=1} \frac{1}{n} \sum_{i=1}^n \left(u^T \left(x_i - \frac{1}{n} \sum_{j=1}^n x_j \right) \right)^2 = \arg \max_{\|u\|=1} u^T C u$$

where

$$C = \frac{1}{n} \sum_{i=1}^n \left(x_i - \frac{1}{n} \sum_{j=1}^n x_j \right) \left(x_i - \frac{1}{n} \sum_{j=1}^n x_j \right)^T = \frac{1}{n} X H X^T, \quad X = [x_1 \dots x_n]^T$$

$$H = I - \frac{1}{n} \mathbf{1}_{n \times n} \quad \text{where } \mathbf{1}_{n \times n} \text{ is a matrix of ones}$$

If $C = Q \Lambda Q^T$, then $u^T C u = (Q^T u)^T \Lambda (Q^T u) = \sum_i \lambda_i \tilde{u}_i^2 \leq \lambda_{\max} \sum_i \tilde{u}_i^2$
 where $\tilde{u} = Q^T u$ are the coordinates of u in the eigenbasis. If $\|u\|^2 = 1$, then $\|\tilde{u}\|^2 = u^T Q Q^T u = u^T u = 1$.

27 Kernel PCA

$$\begin{aligned} f_1 &= \arg \max_{\|f\|_H \leq 1} \frac{1}{n} \sum_{i=1}^n \left(\langle f, \phi(x_i) \rangle_H - \frac{1}{n} \sum_{j=1}^n \langle f, \phi(x_j) \rangle_H \right)^2 \\ &= \arg \max_{\|f\|_H \leq 1} \frac{1}{n} \sum_{i=1}^n \left(f(x_i) - \frac{1}{n} \sum_{j=1}^n f(x_j) \right)^2 = \arg \max_{\|f\|_H \leq 1} \frac{1}{n} \sum_{i=1}^n (f(x_i) - \hat{f}(i))^2 = \arg \max_{\|f\|_H \leq 1} \text{Var}(f) \end{aligned}$$

We can write $f = \sum_{i=1}^n \left(\phi(x_i) - \frac{1}{n} \sum_{j=1}^n \phi(x_j) \right) = \sum_{i=1}^n \alpha_i \tilde{\phi}(x_i)$ since any orthogonal component contributes.

Define infinite dimensional analog of covariance: $C = \frac{1}{n} \sum_{i=1}^n \tilde{\phi}(x_i) \otimes \tilde{\phi}(x_i)$, where $\tilde{\phi}(x_i)$ is demeaned features, $\otimes$ is outer product.

with trick $(A \otimes b)c := \langle b, c \rangle_H a$:

$$Cf^T = \frac{1}{n} \sum_{i,j} \tilde{\phi}(x_i) (\langle x_j, \tilde{\phi}(x_j) \rangle), \quad \tilde{K} = HKH \quad \text{since } \tilde{K} = H^T X^T X H = H^T E H = H E H$$

project $f_i \alpha = C f_i$ and $\tilde{\phi}(x_i) \Rightarrow n \lambda_i \tilde{\alpha}_i = \tilde{K} \tilde{\alpha}_i \Rightarrow n \lambda_i \alpha = K \alpha$

$\|f\|_H^2 = \alpha^T \tilde{K} \alpha = n \lambda \|\alpha\|^2$, so to make it unit norm, we scale $\alpha \in \alpha / \sqrt{n \lambda}$.

28 Ridge Regression

$$X = [x_1, \dots, x_n]^T \in \mathbb{R}^{p \times n}, \quad y := [y_1, \dots, y_n]^T$$

$$\alpha^* = \arg \min_{\alpha} \left(\sum_{i=1}^n (y_i - \alpha^T x_i)^2 + \lambda \|\alpha\|^2 \right) = \arg \min_{\alpha \in \mathbb{R}^p} (\|y - X^T \alpha\|^2 + \lambda \|\alpha\|^2)$$

$$\alpha^* = (X X^T + \lambda I)^{-1} X y = X (X^T X + \lambda I)^{-1} y$$

So solution is in form of $\alpha^* = \sum_{i=1}^n \alpha_i^* x_i$.

Perform SVD on X : $X = U S V^T$, with $U = [u_1, \dots, u_c]$, $S = [\sigma_1, \sigma_2, \dots]^T$, $V = [v_1, \dots, v_n]^T$, $U^T U = U U^T = I_c$, $V^T V = V V^T = I_n$.

29 Kernel Ridge Regression

$$\alpha^* = \arg \min_{\alpha} \left(\sum_{i=1}^n (y_i - \langle \alpha, \phi(x_i) \rangle_H)^2 + \lambda \|\alpha\|_H^2 \right)$$

$$\alpha^* = X (K + \lambda I_n)^{-1} y - \sum_{i=1}^n \alpha_i^* \phi(x_i), \quad \alpha = (K + \lambda I_n)^{-1} y$$

30 Gaussian Kernel

$$k(x, x') = \exp\left(-\frac{\|x - x'\|^2}{\sigma^2}\right)$$

σ , k less centralized, more smoother. $\|f\|_{H_k}^2 = \sum_j \frac{|\hat{f}_j|^2}{\lambda_j}$, so larger k decays faster, f has fat tail become less penalty k !

31 Representer Theorem

Let H be RKHS on a set X with kernel k . $(x_1, \dots, x_n) \subset X$ be training points. Let $L : \mathbb{R}^n \rightarrow \mathbb{R} \cup \{\infty\}$ be any loss that depends only on the vector of point-wise values:

$$L \mapsto (f(x_1), \dots, f(x_n))$$

$\Omega : H_k \rightarrow [0, \infty)$ be a non-decreasing function of RKHS norm, e.g., $\Omega(f) = \lambda \|f\|_{H_k}^2$.

Consider the regularized empirical-risk problem:

$$\min_{f \in H_k} F(f) := L(f(x_1), \dots, f(x_n)) + \Omega(f)$$

Any minimizer of F lies in the finite-dimensional span of the kernel sections at the data points:

$$f(x) = \sum_{i=1}^n \alpha_i k(x, x_i) \quad \text{for some coefficients } (\alpha_1, \dots, \alpha_n) \in \mathbb{R}^n. \text{ If } \Omega \text{ is strictly increasing, } f \text{ is unique}$$

32 Kernel Mean Embedding

We have RKHS H built from kernel $k : X \times X \rightarrow \mathbb{R}$ and a Borel probability measure P on X .

Kernel mean embedding of P :

$$\mu_P := \mathbb{E}_P[k(\cdot, X)] \in H, \quad \text{such that}$$

$$\text{for } \forall f \in H, \quad \mathbb{E}_P[f(X)] = \langle f, \mu_P \rangle_H$$

Define linear functional $T_P : H \rightarrow \mathbb{R}$, $T_P f = \mathbb{E}_P[f(X)]$. With $f(x) = \langle f, \phi(x) \rangle_H$, we have:

$$|T_P f| = |\mathbb{E}_P[f(X)]| \leq \mathbb{E}_P[|f(X)|] = \mathbb{E}_P[|\langle f, k(\cdot, X) \rangle_H|] \leq \mathbb{E}_P[\|f\|_H \|k(\cdot, X)\|_H]$$

If $\mathbb{E}_P[\|k(\cdot, X)\|_H] < \infty$, T_P is bounded.

So $\exists! \mu_P \in H$ such that $T_P f = \langle f, \mu_P \rangle_H$ by Riesz

Let $f = \phi(x) = k(\cdot, x)$, we have:

$$\mu_P(x) = \langle \mu_P, \phi(x) \rangle_H = T_P \phi(x) = \mathbb{E}_P[k(x, X)]$$

μ_P is just the ordinary mean of those feature vectors. $\mu_P = \mathbb{E}_P[\phi(\theta)]$.

$$\langle \mu_P, \mu_Q \rangle_H = \mathbb{E}_P \mathbb{E}_Q [\langle k(\cdot, X), k(\cdot, Y) \rangle_H] = \mathbb{E}_P \mathbb{E}_Q [k(X, Y)]$$

distance between feature means.

33 Empirical Mean Embedding

$$\hat{\mu}_P = \frac{1}{m} \sum_{i=1}^m \phi(X_i), \quad X_i \sim P$$

34 Maximum Mean Discrepancy

$$\text{MMD}^2(P, Q) = \|\mu_P - \mu_Q\|_H^2 = \mathbb{E}_P[\langle (x, X) \rangle] + \mathbb{E}_Q[\langle (y, Y) \rangle]$$

distance between feature means

empirical MMD:

$$\widehat{\text{MMD}}^2 = \frac{1}{n(n-1)} \sum_{i \neq j} k(x_i, x_j) + \frac{1}{n(n-1)} \sum_{j \neq j'} k(y_j, y_{j'}) - \frac{2}{n^2} \sum_i k(x_i, y_i)$$

35 MMD as an Integral Probability Metric

Find a “well learned function” $f(x)$ to maximize $\mathbb{E}_P[\mathbb{E}[f(X)] - \mathbb{E}_Q[f(D)]]$.

$$\text{MMD}(P, Q; F) = \|\mu_P - \mu_Q\|_H^2 = \|T_P - T_Q\|_{F'} = \sup_{\|f\|_H \leq 1} |\langle T_P - T_Q, f \rangle| = \sup_{\|f\|_H \leq 1} |\mathbb{E}_P[f(X)] - \mathbb{E}_Q[f(X)]|$$

F is unit ball in RKHS F .

$$\text{MMD}(P, Q; F) = 0 \Leftrightarrow P = Q.$$

36 Witness Function

Witness function: $f^* \sim \mu_P - \mu_Q$.

$$f^*(v) = \langle f^*, \phi(v) \rangle_H \propto \langle \mu_P - \mu_Q, \phi(v) \rangle_H = \frac{1}{n} \sum_{i=1}^n k(x_i, v) - \frac{1}{n} \sum_{j=1}^n k(y_j, v)$$

37 Hypothesis Testing on MMD

Null hypothesis H_0 : when $P = Q$ should see $\widehat{\text{MMD}}^2$ close to 0. When $P \neq Q$, statistic is asymptotically normal:

$$\frac{\widehat{\text{MMD}}^2 - \text{MMD}^2(P, Q)}{\sqrt{V_n(p, Q)}} \xrightarrow{D} \mathcal{N}(0, 1)$$

when $P = Q$, statistic has asymptotic distribution $n\widehat{\text{MMD}}^2 \sim \sum_{\ell=1}^{\infty} \lambda_{\ell} [\mathbb{Z}_{\ell}^2 - 2]$
 $\mathbb{Z}_{\ell} \sim \mathcal{N}(0, 2)$ i.i.d.

We can permute samples from P and Q to simulate $P = Q$ and get α (false positive rate bound).

38 Maximum Mean Embedding via Fourier Series

$$\mu_P(x) = \langle \mu_P, k(\cdot, x) \rangle_H = \mathbb{E}_P[\mathbb{E}[k(t - x)]] = \int_X k(t - x) dP(t)$$

$$\text{MMD}(P, Q; H(\mathcal{K})) = \|\mu_P - \mu_Q\|_H^2 = \left\| \sum_{\ell=0}^{\infty} [\phi_{P,\ell} - \phi_{Q,\ell}] \mathcal{K}_{\ell}^2 \exp(i\omega_{\ell} x) \right\|_H^2,$$

$$\text{MMD}^c(P, Q; H^f) = \sum_{\ell=0}^{\infty} |\varphi_{P,\ell} - \varphi_{Q,\ell}|^2 \mathcal{K}_{\ell}^2 = \sum_{\ell=0}^{\infty} |\varphi_{P,\ell} - \varphi_{Q,\ell}|^2 \mathcal{K}_{\ell}$$

$$\mathcal{K}(x, y) = k(x - y) = \sum_{\ell=0}^{\infty} \mathcal{K}_{\ell} e^{i\omega_{\ell}(x-y)}, \mathcal{K}_{\ell} > 0$$

$$\psi_{\rho,\ell} := \int e^{i\ell t} dP(t) \text{ (characteristic function).}$$

$$\mu_P(x) = \int_X k(t - x) dP(x) = \int_X \sum_{\ell} \mathcal{K}_{\ell} e^{i\omega_{\ell}(t-x)} dP(t) = \sum_{\ell} \mathcal{K}_{\ell} e^{i\omega_{\ell} x} \int_X e^{-i\omega_{\ell} t} dP(t) = \sum_{\ell} \mathcal{K}_{\ell} e^{i\omega_{\ell} x} \psi_{\rho,\ell}$$

(we can use $e^{i\omega x}$ since $\mathcal{K}_{\ell} = \mathcal{K}_{-\ell}$)

$$|\psi_{\rho,\ell} - \hat{\varphi}_{\ell}|^2 = |\varphi_{\rho,\ell} - \varphi_{\ell+m}|^2 \text{ since } [\hat{\mathcal{K}}_{\ell}^2 = 2\tilde{\mathcal{K}}.$$

39 Kernel is Characteristic

A kernel is characteristic if the mean-embedding map $P \mapsto \mu_P := \int k(\cdot, x)dP(x) \in H_k$ is injective, i.e., $\mu_P = \mu_Q \Rightarrow P = Q$.

For translation-invariant kernel $\mathcal{K}(x, y) = k(x - y)$, $\mu_P = \int k(\cdot - x)dP(x) \in H_k$.

μ_P is convolution of $\mathcal{K} * P$, by convolution theorem, $\widehat{\mathcal{K} * P}(\omega) = \hat{\mathcal{K}}(\omega)\hat{P}(\omega)$, where $\mathcal{K}(t) = (\int e^{-i\omega x t} d\Lambda(\omega))$, $\hat{P}(\omega) = (\int e^{-i\omega x} dP(x))$.

so $\mu_P = \mu_Q \Leftrightarrow \hat{\mathcal{K}}(\omega)[\hat{P}(\omega) - \hat{Q}(\omega)] = 0 \ \forall \omega$.

So full-support spectrum $\hat{\mathcal{K}}(\omega) \neq 0 \ \forall \omega$ forces $\hat{P} = \hat{Q}$ everywhere, so $P = Q$.

In continuous case, $\mathcal{K}(t) = \int e^{i\omega x t} d\Lambda(\omega)$, $\hat{P}(\omega) = \int e^{-i\omega x} d\Lambda(\omega)$, $\hat{P}(\omega) = \frac{d\Lambda(\omega)}{d\omega}$.

In periodic domain case, spectrum is a sum $\{\lambda_\ell\}$ (continuous)

so kernel is characteristic $\Leftrightarrow \{\}$

40 Fourier Representation of MMD on $\mathbb{R}^d$

$$\text{MMD}^2(P, Q; F) = \int |\hat{P}(\omega) - \hat{Q}(\omega)|^2 d\Lambda(\omega)$$

41 Properties of MMD and Kernels

1. **MMD decay:** with increasing perturbation frequency as $\mathcal{K}$ decays as ℓ grows.
2. **Translation invariant kernel:** k is characteristic for probability measures on $\mathbb{R}^d \Leftrightarrow \text{supp}(\Lambda) = \mathbb{R}^d$ (support zero on at most a countable set).
3. **Any continuous compactly supported kernel:** is characteristic (since Fourier spectrum $\Lambda(\omega)$ cannot be 0 on an interval).
4. **Universal RKHS:** $k(x, x')$ continuous, X compact, and F dense in $C(X)$ with respect to $\|\cdot\|_\infty$.

If X is compact, continuous $k(\cdot, x)$ creates compact-image.

If X is not compact, $\|k(\cdot, x)\|_F < \infty$.

F is dense in $C(X) \Rightarrow$ under $\|\cdot\|_\infty \Leftrightarrow \forall g \in C(X), \forall \varepsilon > 0, \exists f \in F$ such that $\|f - g\|_\infty < \varepsilon$.

5. **Universal $F \Rightarrow$ kernel is characteristic**

42 Test Power and Kernel Selection

Test power depends on $k(x, x')$, P and Q (and n), with characteristic kernel, MMD has power $\rightarrow 1$ as $n \rightarrow \infty$ for any fixed problem. But could be terrible for reasonable n .

$$\mathbb{P}[\widehat{n\text{MMD}}^2 \geq \xi] \Rightarrow \mathbb{E} \left(\frac{\widehat{\text{MMD}}(P, Q) - c_n}{\sqrt{V_n(p, Q)}} \right), \quad \text{second term is } O(\hat{n})$$

to maximize test power, maximize $\frac{\widehat{\text{MMD}}(P, Q)}{\sqrt{V_n(p, Q)}}$ for fix n .

Split data first to choose a kernel k maximizing the term and use the k to perform MMD test on rest of data.

43 Type I and Type II Errors

1. **Type I error:** false positive $\rightarrow$ reject a true H_0
2. **Type II error:** false negative $\rightarrow$ accept a false H_0

44 MMD as a Dependence Measure

$$\text{MMD}[\rho_{XY}, P_X P_Y, H_k] = \text{Simulate } Q \text{ with pairs } (x_i, y_j)$$

45 Constrained Covariance

$$\text{COCO}(\rho_{XY}) = \sup_{\|f\|_F=1, \|g\|_G=1} \text{cov}[f(x)g(y)], \quad f \in F, g \in G, \text{ two spaces}$$

COCO = max singular value of feature covariance $C(\rho(f, g))$.

46 Hilbert-Schmidt Operators

Let F and G be separable Hilbert spaces. $\{e_i\}_{i \in J}$ are orthonormal basis of F , $\{f_j\}_{j \in I}$ for G .

Define two compact linear operators $L : G \rightarrow F$ and $M : G \rightarrow F$:

$$\|L\|_{HS}^2 = \sum_{j \in J} \|Lf_j\|_F^2 = \sum_{i \in J} \sum_{j \in I} |\langle Lf_j, e_i \rangle_F|^2$$

Hilbert-Schmidt operators form a Hilbert space $HS(G, F)$ with inner-product:

$$\langle L, M \rangle_{HS} = \sum_{j \in J} \langle Lf_j, Mf_j \rangle_F = \sum_{i \in I} \sum_{j \in J} \langle Lf_j, e_i \rangle_F \langle Mf_j, e_i \rangle_F$$

47 Finite Dimensions and Frobenius Norm

In finite dimensions, if L and M are matrices, the HS inner product reduces to the Frobenius inner product:

$$\langle L, M \rangle_{HS} = \text{tr}(L^T M), \quad \|L\|_{HS}^2 = \sqrt{\text{tr}(L^T L)} = \sum_{i,j} |L_{ij}|^2, \quad \hat{\mu}_i := \frac{1}{n} \sum_{i=1}^n \phi(x_i)$$

Hilbert-Schmidt inner product measures similarity of L and M on basis functions.

If L and M transform OMS in a similar way, their HS inner product is large.

48 Rank-One Operator

A rank-one operator $T : G \rightarrow \mathbb{R}F$ can be written as:

$$T = a \otimes b, \quad a \in F, b \in G$$

For $\psi \in G$, $(a \otimes b)\psi = \langle b, \psi \rangle_G a$.

The image of T is the one-dimensional subspace $\text{span}\{a\}$, hence rank-one:

$$\|a \otimes b\|_{HS}^2 = \sum_{j \in J} \|(a \otimes b)f_j\|_F^2 = \sum_{j \in J} |\langle b, f_j \rangle_G|^2 \|a\|_F^2 = \|a\|_F^2 \sum_{j \in J} \|b, f_j\|_G^2 = \|a\|_F^2 \|b\|_G^2 < \infty$$

$a \otimes b$ is a Hilbert-Schmidt operator.

49 Expected Hilbert-Schmidt Inner Product Functional

Define linear functionals operator by expected Hilbert-Schmidt inner product functional:

$$T_{xy} : HS(G, F) \rightarrow \mathbb{R}$$

$$T_{xy}(A) = \mathbb{E}_{xy} [\langle \phi(x) \otimes \psi(y), A \rangle_{HS}]$$

It is a bounded operator if:

$$\mathbb{E}_{xy} \|\phi(x) \otimes \psi(y)\|_{HS} = \mathbb{E}_{xy} [\|\phi(x) \otimes \psi(y)\|^2] < \infty$$

then $T_{xy} \in HS^*$ (a continuous dual of HS)

and by Riesz representation theorem: $\exists \hat{C}_{xy} \in HS(G, F)$ such that:

$$T_{xy}(A) = \langle \hat{C}_{xy}, A \rangle_{HS}$$

50 Rank-One Covariance Operator

We choose $A = f \otimes g$. Then:

$$\langle \hat{C}_{xy}, f \otimes g \rangle_{HS} = \langle f, \hat{C}_{xy}g \rangle_F = \mathbb{E}_{xy} [\langle \phi(x), f \rangle_F \langle \psi(y), g \rangle_F]$$

Define by expected operator:

$$T_{xy} : HS(G, F) \rightarrow \mathbb{R}, \quad T_{xy}(A) = \mathbb{E}_{xy} [\langle \phi(x) \otimes \psi(y), A \rangle_{HS}]$$

51 Empirical Covariance Operator

$$\hat{C}_{xy} = \frac{1}{n} \sum_{i=1}^n \phi(x_i) \otimes \psi(y_i) - \mu_x \otimes \mu_y, \quad \text{where} \quad \mu_x := \frac{1}{n} \sum_{i=1}^n \phi(x_i)$$

$$\hat{C}_{xy} = \frac{1}{n} X^T Y, \quad X = [\phi(x_1), \dots, \phi(x_n)]^T, \quad Y = [\psi(y_1), \dots, \psi(y_n)]^T$$

$$\tilde{C}_{xy} = H R H, \quad \tilde{C}_{xy} = H L H$$

$$\text{COCO}(P_{xy}) = \sup_{\|f\|=1, \|g\|=1} \text{cov}[f(x)g(y)]$$

We can do maximize $\langle f, \hat{C}_{xy}g \rangle_F$ subject to $\|f\|_F = 1, \|g\|_G = 1$. Witness function should be $f = \frac{1}{\sqrt{n}} \sum_i a_i [\phi(x_i) - \mu_x]$ and $g = Y H P$.

$$f \hat{C}_{xy} g = \frac{1}{n} \alpha^T K L \beta, \text{ and } \|f\|_F^2 = \alpha^T K \alpha, \|g\|_G^2 = \beta^T L \beta.$$

52 Associated Lagrangian

$$\text{COCO}(P_{xy}) = \sup_{\|f\|=1, \|g\|_G=1} \text{cov}[f(x)g(y)]$$

With Lagrangian formula with multipliers $\gamma \in \mathbb{R}^m$:

$$\mathcal{L}(P, Q; F, g) = -f^T \hat{C}_{xy} g + \frac{1}{2}[\|f\|_F^2 - 1] + \frac{1}{2}c[\|g\|_G^2 - 1]$$

where $\alpha_0 = \sup(\mathbb{E}_{xy}[\phi(x) \otimes \psi(y)]\|_{HS})$, $\alpha = r$.

53 Hilbert-Schmidt Independence Criterion (HSIC)

$$\text{HSIC}^2(\mathcal{P}_X, \mathcal{P}_Y; F, G) = \sum_{i=1}^{\infty} \delta_i^2, \quad \delta_i := \text{CoCD}(\mathcal{P}_{XY}; F, G)$$

HSIC is MMD with product kernel

$$\text{HSIC}(P_{XY}; F, G) = \text{MMD}(\mathcal{P}_X, P_X P_Y; H_k) \quad \text{where } k((x, y), (x', y')) = k(x, x')l(y, y')$$

$$\text{HSC}^2(P_{XY}; F, G) := \|\hat{C}_{XY} - \mu_X \otimes \mu_Y\|_{HS}$$

$$= \langle \hat{C}_{XY}, \hat{C}_{XY} \rangle_{HS} + \langle \mu_X \otimes \mu_Y, \mu_X \otimes \mu_Y \rangle_{HS} - 2\langle \hat{C}_{XY}, \mu_X \otimes \mu_Y \rangle_{HS}$$

$$\|\hat{C}_{XY}\|_{HS}^2 = \mathbb{E}_X[k(x, x')]\mathbb{E}_Y[\ell(y, y')] = D \cdot B$$

$$\|\mu_X \otimes \mu_Y\|_{HS}^2 = \|\mu_X\|_F^2 \|\mu_Y\|_G^2 = \mathbb{E}_{XY}[k(x, x')]\mathbb{E}_Y[\ell(y, y')] =: B$$

$$\langle \hat{C}_{XY}, \mu_X \otimes \mu_Y \rangle_{HS} = \mathbb{E}_{XY}[\langle \phi(x) \otimes \psi(y), \mu_X \otimes \mu_Y \rangle_{HS}] = B$$

So $\text{HSIC}^2 = A - 2B + D$

54 Unbiased and Biased HSIC Estimators

Unbiased $\hat{A} := \frac{1}{n(n-3)} \sum_{i=1}^n \sum_{j \neq i, j'} k(x_i, x_j) \ell(y_i, y_j)$.

Biased $\hat{A}_b := \langle \frac{1}{n} \sum_i \phi(x_i) \otimes \psi(y_i), \frac{1}{n} \sum_i \phi(x_i) \otimes \psi(y_i) \rangle_{HS}$

$$= \frac{1}{n^2} \sum_{i,j} k(x_i, x_j) \ell(y_i, y_j) = \frac{1}{n^2} \text{tr}(KL).$$

Expectation of bias is of $O(\frac{1}{n})$.

55 Empirical HSIC

$$\text{HSIC}^2 = \frac{1}{n^2} \text{tr}(KLH), \quad \text{HSIC}^2(Q) = \sum_{i=1}^{\infty} \lambda_i z_i^2, \quad z_i \sim \mathcal{N}(0, 1) \text{ i.i.d.}$$

56 Asymptotics of HSIC

when $P_{XY} = P_X P_Y P^*$: $n\text{HSIC}^2 \Rightarrow \sum_{\ell=1}^{\infty} \lambda_{\ell} z_{\ell}^2$, $z_{\ell} \sim \mathcal{N}(0, 1)$ i.i.d.

57 Null Distribution

Given $P_{XY} = P_X P_Y$, what is threshold C_{α} such that $\mathbb{P}(\text{HSIC}^2 > C_{\alpha}) < \alpha$?

Under null distribution via permutation: permute y and calculate HSIC many times to get empirical CDF.

Under null distribution via moment matching:

$$n\text{HSIC}^2(\hat{Z}) \sim \frac{\alpha^2}{n^2} T(\alpha), \quad \alpha = \frac{\mathbb{E}(\text{HSIC}^2)^2}{\text{Var}(\text{HSIC}^2)}, \quad \beta = \frac{n\mathbb{E}(\text{HSIC}^2)}{C_{\alpha}}$$

58 V-structure Discrepancy

CONSISTS CI test: $H_0 : X \perp Y | Z$ **Factorization test:** $H_0 : (X, Y) \perp Z | TV(Y, Z) \perp X$.

59 Lancaster Interaction Measure

Lancaster interaction measure of $(X_1, \dots, X_D) \wedge P$ is a signed measure ΔP that vanishes whenever P can be factorised non-trivially.

$$D = 2 : \Delta P = P_{XY} - P_X P_Y, \quad D = 3 : \Delta P = P_{X_1 X_2 X_3} - P_{X_1} P_{X_2 X_3} - P_{X_2} P_{X_1 X_3} - P_{X_3} P_{X_1 X_2} + 2P_{X_1} P_{X_2} P_{X_3}$$

$$(X, Y) \perp Z \Rightarrow \Delta P = 0$$

but $\Delta_1 P = 0 \Rightarrow (X, Y) \perp Z \vee (X, Z) \perp Y \vee (Y, Z) \perp X$. It is only useful when $\Delta P \neq D$.

60 HSIC on Feature Selection

Backwards elimination of irrelevance features to maximize HSIC.

61 Kernel Stein Discrepancy (KSD)

$$\begin{aligned} \text{KSD}(\pi) &:= \sup_{\|g\|_{\ell_{\infty}} \leq 1} \mathbb{E}_{\pi}[[\nabla_x \log \pi(x^*)]] \\ &= \sup_{\|g\|_{\ell_{\infty}} \leq 1} \mathbb{E}_{\pi}[\nabla_x \log p(x)] \end{aligned}$$

62 Langevin Stein Operator

$$\mathcal{A}_p^f(x) = \frac{1}{p(x)} \frac{d}{dx} [f(x)p(x)] = \nabla_x \log p(x) \cdot f(x) + \nabla_x f(x)$$

$$\mathbb{E}_p[\mathcal{A}_p^f] = 0 \quad \text{subject to} \quad [f(x)p(x)]_\infty = 0$$

Here we do not need to normalize $p(x)$.

63 Kernel Stein Discrepancy (KSD)

$$\begin{aligned} \text{KSD}_p(\mathcal{Q}) &= \sup_{\|g\|_F \leq 1} \mathbb{E}_Q[\mathcal{A}_p g] \\ &= \sup_{\|g\|_F \leq 1} \mathbb{E}_Q[g]. \end{aligned}$$

64 Reproducing Property for the Derivative

$$\frac{d}{dx} f(x) = \left\langle f, \frac{d}{dx} k(x, x') \right\rangle_F, \quad \left\langle \frac{d}{dx} \phi(x), \psi(x) \right\rangle_F = \frac{d}{dx} k(x, x)$$

65 Derivatives of Functions

$$\begin{aligned} [\mathcal{A}_p f](x) &= \left(\frac{d}{dx} \log p(x) \right) f(x) + \frac{d}{dx} f(x) = \left\langle f, \frac{d}{dx} k(x, x') \right\rangle_F \\ z_\pi(x) &= \left(\frac{d}{dx} \log p(x) \right) (\psi(x)) + \frac{d}{dx} \psi(x) \end{aligned}$$

66 Expected Stein Operator

$$\begin{aligned} T_{\rho, p^*} &= \mathbb{E}_Q[\mathcal{A}_p f(x)] = \left\langle f, \mathbb{E}_Q \left[\frac{d}{dx} \log p(x) \right] \psi(x) + \frac{d}{dx} \psi(x) \right\rangle_F \\ (T_{\rho, p} : H &\rightarrow \mathbb{R}) \end{aligned}$$

$$|\mathbb{E}_Q[\langle f, z_x \rangle_F]| \leq \mathbb{E}_Q[|\langle f, z_x \rangle_F|] \leq \|f\|_F \mathbb{E}_Q[\|z_x\|_F]$$

$$\|z_x\|_F^2 = C + \left(\frac{d}{dx} \log p(x) \right)^2 c, \quad \text{for boundedness we need to have } \mathbb{E}_Q \left[\left(\frac{d}{dx} \log p(x) \right)^2 \right] < \infty$$

So that $\mathbb{E}_Q[\|z_x\|_F] = \mathbb{E}_Q \left[C + \left(\frac{d}{dx} \log p(x) \right)^2 c \right] \leq \sqrt{\mathbb{E}_Q \left[c + \frac{d}{dx} \log p(x) \right]^2} c < \infty$.

If data is heavy-tailed and model has right tail, KSD may not be defined.

So by Riesz, $T_{\rho, p^*} = \mathbb{E}_Q[\mathcal{A}_p f(x)] = \langle f, z_x \rangle_F$.

67 KSD Properties

$$\text{KSD} = \sup_{\|g\|_F \leq 1} \mathbb{E}_Q[\mathcal{A}_p g] = \sup_{\|g\|_F \leq 1} \mathbb{E}_Q[g], \quad \mathbb{E}_Q[z_x] \sim \mathcal{F}$$

witness function should be $g \propto \mathbb{E}_Q[z_x]$.

68 Squared KSD

KSD can be written as $\text{KSD}^2(\mathcal{Q}, P) = \mathbb{E}_{x, x' \sim Q}[k_p(x, x')]$, where $k_p(x, \ell) = \mathcal{A}_p^x \mathcal{A}_p^y k(x, x')$, and $\mathcal{A}_p^{(1)}, \mathcal{A}_p^{(2)}$ act on the first and second arguments of k .

$$\mathcal{K}_p(x, y) = S_p(x)^T S_p(y) k(x, y) + S_p(x)^T k_2(x, y) + k_1(x, y) k(x, y) + \sigma[k_{\alpha_1}(x, y)]$$

$$S_p(x) = \frac{\nabla p(x)}{p(x)}, \quad k_1(x, x') = \nabla_x k(x, x'), \quad k_2(x, x') = \nabla_y k(x, x'), \quad h_p(x, x') = \nabla_x \nabla_y k(x, x')$$

69 Estimating KSD

$$\text{KSD}(Q, P; F) = \frac{1}{n(n-1)} \sum_{i=1}^n \sum_{j \neq i} \delta_i \delta_j h_p(z_i, z_j), \quad \text{where } \{k_i\}_{i=1}^n \text{ i.i.d., } \mathbb{E}(\delta_i) = 0 \text{ and } \mathbb{E}(\delta_i^2) = 1$$

$$\sqrt{n} \left(\text{KSD}_p(\hat{Q}) - \text{KSD}_p(Q) \right) \xrightarrow{d} \mathcal{N}(0, \delta_p^2) \quad \text{where } \delta_p^2 = 4\text{Var}(\mathbb{E}_x[\mathcal{A}_p \ell(y, x')])$$

For latent variable model testing $\hat{p}(x) = \int p(x|z)p(z)dz$, $p(x)$ is usually not tractable. we use $S_p(x) = \frac{\nabla p(x)}{p(x)} = \frac{\nabla p(x|z)}{p(x)} = \int \frac{\nabla p(x|z)}{p(x|z)} p(z|x) dz = \mathbb{E}_{z|x}[S_p(x|z)]$.

70 Comparing Models via Samples

Compare sample from Q and a model P: usually can't compute $\mathbb{E}_P$ in closed form.

Compare a sample and a model: KSD.

71 Hyperplane in $\mathbb{R}^D$

For a hyperplane $\{x \in \mathbb{R}^D : w^T x + b = 0\}$, w is perpendicular to the hyperplane.

72 Distance of Point to Hyperplane

Distance of a point x to the hyperplane is $\frac{|w^T x + b|}{\|w\|}$.

73 Generic Optimization Problem

Generic optimization problem on $X \in \mathbb{R}^D$:

$$\begin{aligned} & \text{minimize} && f_0(x) \\ & \text{subject to} && f_i(x) \leq 0, \quad i = 1, \dots, m \\ & && h_i(x) = 0, \quad i = 1, \dots, p \end{aligned}$$

Lagrangian L is a lower bound on the original problem.

$$L(x, \lambda, \nu) := f_0(x) + \sum_{i=1}^m \lambda_i f_i(x) + \sum_{i=1}^p \nu_i h_i(x)$$

$$I_-(w) = \begin{cases} 0 & w \leq 0 \\ \infty & w > 0 \end{cases}, \quad I_0(w) = \begin{cases} 0 & w = 0 \\ \infty & w \neq 0 \end{cases}$$

To ensure a lower bound, $\lambda \geq 0$.

$$\inf_x g(x, \lambda, \nu) = \inf_x L(x, \lambda, \nu) \leq p^*$$

when $\lambda \geq 0$ and $f_i(x) \leq 0$, Lagrange penalizes out of bounds of f_i and reward is the bound as $\lambda_i f_i(x) > 0$ when out of bound while $\lambda_i f_i(x) \leq 0$ when in the bound.

74 Lagrange Dual Problem

Lagrange dual problem: maximize $g(\lambda, \nu)$ subject to $\lambda \geq 0$.

Dual function is a pointwise infimum of affine functions of (λ, ν) with different x , hence is concave in (λ, ν) with convex constraint set $\lambda \geq 0$.

$$d^* = \max g(\lambda, \nu) \leq p^*$$

Strong duality means $d^* = p^*$.

75 Strong Duality Condition

Strong duality holds if:

1. the primal problem is convex, i.e., $f_0, \dots, f_m$ are convex minimize $f_0(x)$ subject to $f_i(x) \leq 0 \quad i = 1, \dots, m, Ax = b$.
2. **Slater's condition holds:** there exists some strictly feasible point $x^* \in \text{relint}(\mathcal{D})$ such that $f_i(x^*) < 0, i = 1, \dots, m, Ax^* = b$.
3. In affine f_i case, we allow $f_i(x^*) \leq 0$. (Only need strict inequality over non-affine constraints).

76 KKT Conditions

Under strong duality, $x^* \in \arg \max_x \min_y L(x, y)$ and y^* are primal and dual solutions if:

1. **Stationarity:** $0 \in \partial f_0(x^*) + \sum_i \lambda_i^* \partial f_i(x^*) + \sum_j \nu_j^* h_j(x^*)$
2. **Complementary slackness:** $\lambda_i^* f_i(x^*) = 0, i = 1, \dots, m.$
3. **Primal feasibility:** $f_i(x^*) \leq 0, i = 1, \dots, m, h_i(x^*) = 0, i = 1, \dots, p.$
4. **Dual feasibility:** $\lambda_i^* \geq 0, i = 1, \dots, m.$

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This is a necessary condition for optimality:

$$\lambda_i^* f_i(x^*) = 0, \quad \text{which means} \quad \lambda_i^* > 0 \Rightarrow f_i(x^*) = 0$$

$$f_i(x^*) < 0 \Rightarrow \lambda_i^* = 0$$

In other f_i case, we allow $f_i(x^*) \leq 0$. (Only need strict inequality over non-affine constraints).